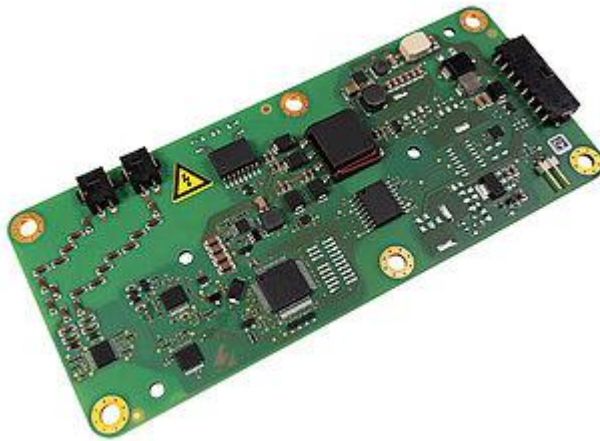


Isolation Monitoring Interrupter (IMI) Testing, Verification and Comms

Type

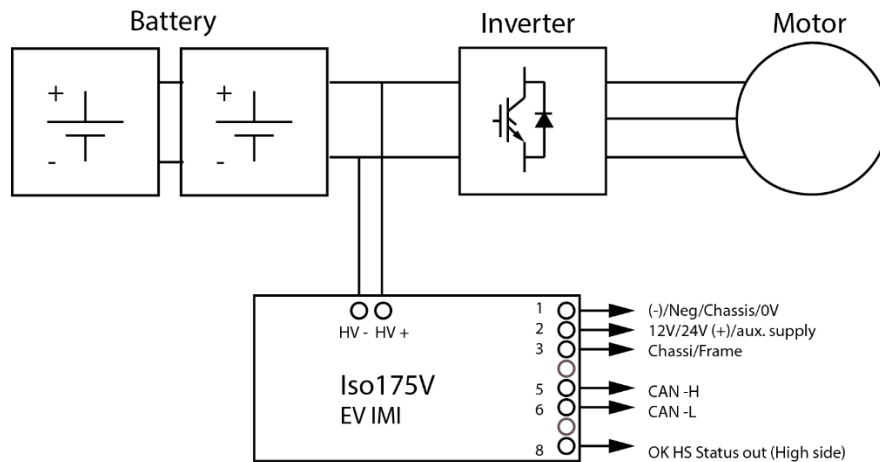
Iso175C



Features

- ☐ Suitable for 12V and 24VDC aux. supply
- ☐ Monitoring insulation in HV systems up to 1000VDC
- ☐ Integrated self-test
- ☐ IMI measurement from 0 ... 50M Ω
- ☐ Speed start measured result after 5 sec
- ☐ Accurate result after 20 sec
- ☐ y-Cap capable up to 10 μ F
- ☐ Interfaces PWM and CAN (HS-CAN & SAE J1939)
- ☐ De-couple function via CAN command

Basic Connection



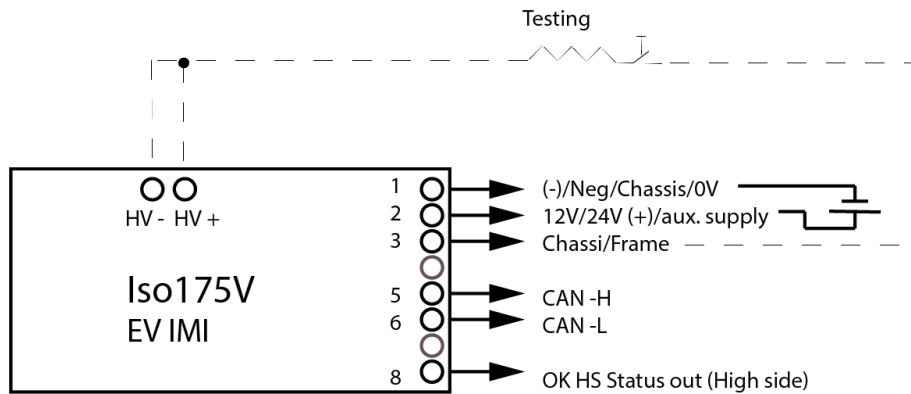
Functionality

The Iso175 is a modern Isolation Monitor Interrupter (IMI) designed to constantly check for isolation between a High voltage (HV) system and a frame/chassis of a battery or fuel cell driven vehicle. It does so by superimposing a small measuring signal onto the HV system. The signal will enter the HV by means of hard-wired pilot connections (HV+, HV-), penetrate all electrically connected devices and attempt to breakthrough to the frame/chassis. The signal will then return via the frame connection (Pin3) into the IMI and be evaluated for signal strength. The strength corresponds to an insulation value in Ohms. There are two means available to alert the user that an ohmic breakdown has occurred. A) By analyzing the CAN message (Pin 5,6) and B) By observing a status change on the OKHs driver (Pin 8). The threshold has to be set via the CAN bus or may come preset by Bender. Generally accepted is a value of 500Ohm/Volt. If a system has 700V max, then the resulting threshold that needs to be selected for alarm shall be $500 \times 700 = 350\text{KOhm}$.

Rules

To comply with Formula SAE regulations (specifically Rule GR 4.3.2 and the IEC 61557-8 standard), an insulation monitoring device (IMD) must include a dedicated hardware fault output capable of directly alerting in the event of an insulation fault. This ensures that fault response is independent of any software, communication protocols, or controller interpretation, allowing for immediate and deterministic shutdown in critical situations. Software-based or communication-only signaling methods, such as CAN or PWM alone, do not meet these requirements. Only IMDs with a true hardwired output—typically an open-drain signal or relay contact—are accepted. Teams must ensure that any IMD used provides this functionality and should seek formal approval before integration into the vehicle.

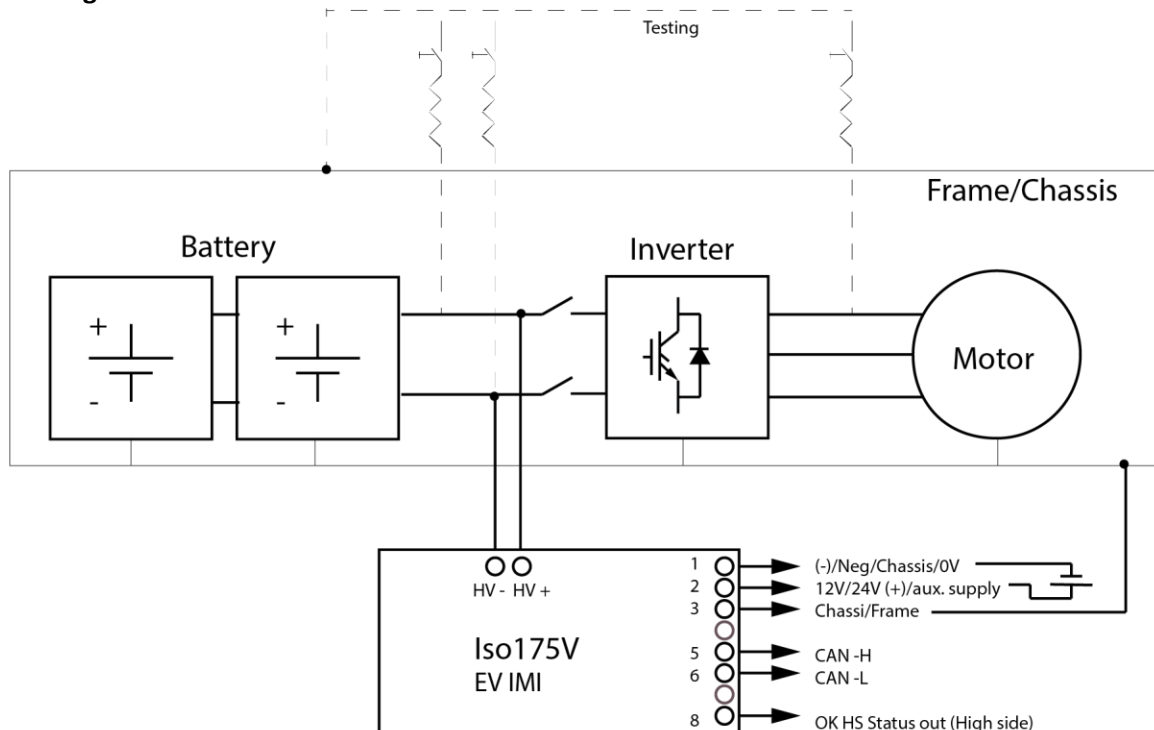
Bench testing



Note: For bench or “dry” testing, the HV+ and HV- connections must be bridged. The IMI expects a low ohm connection between these terminals since in “real” applications there’s usually a HV battery present. This must be simulated by employing a bridge, otherwise the IMIs calculations are offset by 1.2M Ω and a connection loss alarm may occur.

Testing: Connect the device as shown, power the IMI with 12VDC and vary the resistance above and below the threshold. Record the responses on CAN and OKHs

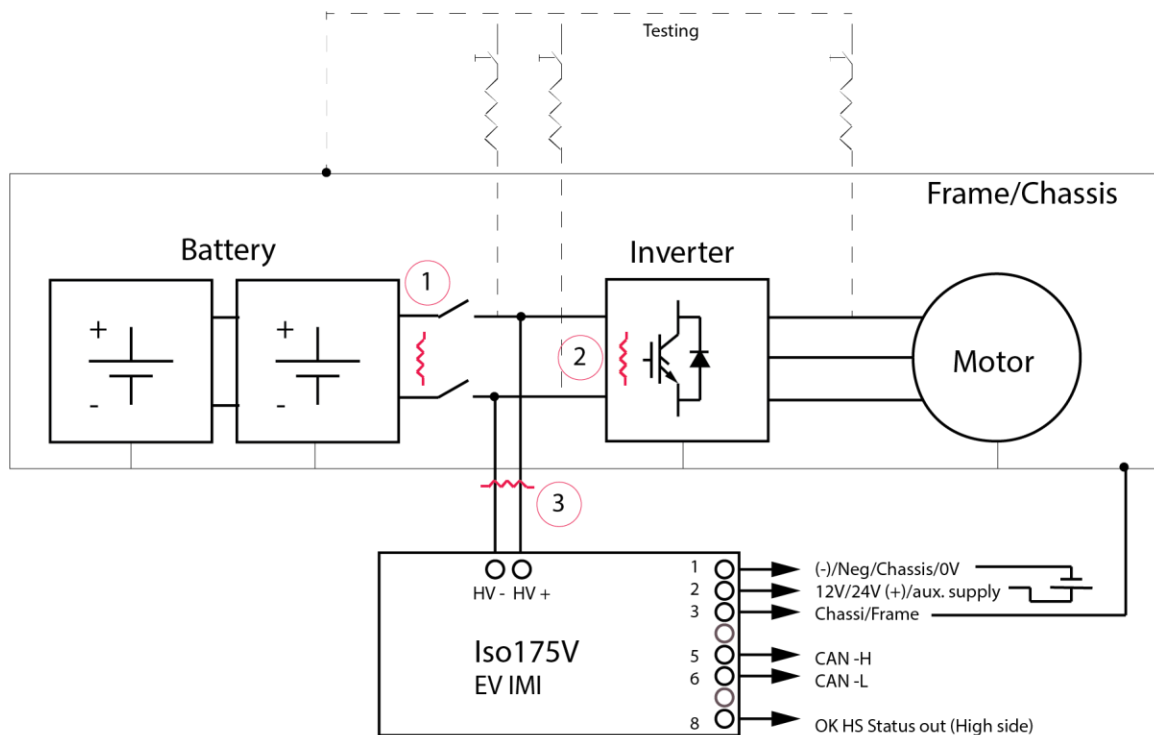
Testing on a vehicle



Note: The schematic above displays an entire drive train. The IMI cannot measure through isolation transformers, open switches, disabled inverters. It can monitor however through an active, conduction inverter. In an operating vehicle, the IMI will measure the entire vehicle from the battery to the motor.
 Note: IMI connections shown on the battery side.

Testing: Connect the device as shown, power the vehicle, power the IMI with 12VDC and vary the resistance above and below the threshold. Record the responses on CAN and OKHs

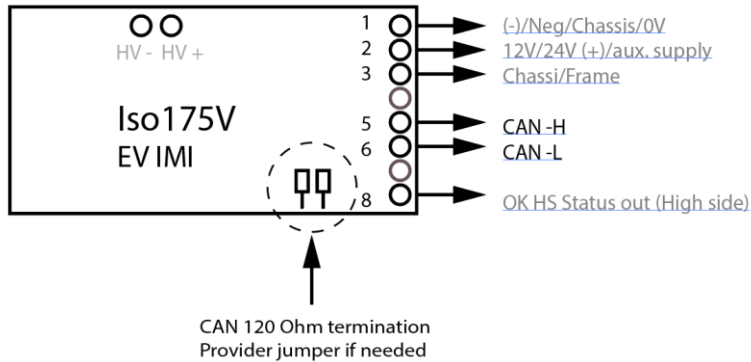
Testing on a vehicle special consideration



Note that in the schematic above the pilot wires now have been moved to the right-hand side of the switches. (1) in this configuration the IMI encounters an open switch on the left (1) and a non-conducting inverter on the right (2). This results in an open circuit between HV+ and HV- (3) and we encounter the same situation as in the test bench scenario above. The readings of the IMI will be off by a magnitude and a connection loss alarm can occur. To rectify the situation, power has to be applied, or a low ohm connection has to be established.

Using the CAN SAEJ1939 Interface

The Iso175 uses a SAEJ1939 CAN based system to communicate with external devices. For these Pin 5 and 6 representing CAN High and Can Low must be used. The Baud rate is 500KB. There is an additional set of PINs available to provide a 120 Ohm termination via jumper if needed. Please note that the little-endian format is used for the value conversions.



Command list

The following may be used as an example for typical commands that can be used with an Iso175

Columns descriptions:

; +Message ID

; | +Cycle time in ms (0=manual)

; | | +Data length

; | | | +Frame type

; | | | | +Message data

; | | | |

18EAF417h 100 3 D 00h FFh 00h Paused ; Request Device Info

18EAF417h 100 3 D 01h FFh 00h Paused ; Request Info General

18EAF417h 100 3 D 02h FFh 00h Paused ; Request Isolation Detail

18EAF417h 100 3 D 03h FFh 00h Paused ; Request Info Voltage

18EAF417h 100 3 D 04h FFh 00h Paused ; Request It-System info

18EFF417h 0 1 D 16h ; Read HW ID Article Number A

18EFF417h 0 1 D 2Ah ; Read Unbalance

18EFF417h 0 1 D 2Eh ; Read Unbalance Threshold

18EFF417h 0 1 D 34h ; Read Device Error

18EFF417h 0 1 D 38h ; Read IsoProfile_Runtime

18EFF417h 0 1 D 3Ah ; Read IsoProfile_Startup

18EFF417h 0 1 D 3Eh ; Read Isulation Quality

18EFF417h 0 1 D 40h ; Read Isolation R-iso neg

18EFF417h 0 1 D 42h ; Read Isolation R-iso pos

18EFF417h 0 1 D 44h ; Read Isolation R-iso Status

18EFF417h 0 1 D 46h ; Read Isolation Threshold_Error

18EFF417h	0 1 D 48h ; Read Isolation Threshold_Timeout_Measurement
18EFF417h	0 1 D 4Ah ; Read Isolation Threshold_Warning
18EFF417h	0 1 D 4Ch ; Read Isolation R_iso_corrected
18EFF417h	0 1 D 4Eh ; Read Isolation R_iso_original
18EFF417h	0 1 D 50h ; Read Isolation Time_elapsed_since_last_measurement
18EFF417h	0 1 D 52h ; Read Capacity
18EFF417h	0 1 D 58h ; Read Selftest Period
18EFF417h	0 1 D 5Ah ; Read Frequency
18EFF417h	0 1 D 5Eh ; Read Voltage HV
18EFF417h	0 1 D 60h ; Read Voltage HVneg – Earth
18EFF417h	0 1 D 62h ; Read Voltage HVpos – Earth
18EFF417h	0 1 D 64h ; Read VoltMode
18EFF417h	0 1 D 66h ; Read VoltThreshold
18EFF417h	0 1 D 68h ; Read DeviceActivity
18EFF417h	0 1 D 6Ah ; Read LockState
18EFF417h	0 1 D 6Ch ; Read Warnings_Alarms
18EFF417h	0 1 D 70h ; Read EarthLift
18EFF417h	0 2 D 2Fh 00h ; .Write Unbalance Threshold = 0 (off)
18EFF417h	0 2 D 2Fh 05h ; .Write Unbalance Threshold = 5%
18EFF417h	0 2 D 2Fh 1Eh ; .Write Unbalance Threshold = 30%
18EFF417h	0 2 D 39h 01h ; . Write IsoProfile_Runtime = StandardFastStartup (1)
18EFF417h	0 2 D 39h 02h ; . Write IsoProfile_Runtime = Standard (2)
18EFF417h	0 2 D 39h 03h ; . Write IsoProfile_Runtime = HighCapacityFastStartup (3)
18EFF417h	0 2 D 39h 04h ; . Write IsoProfile_Runtime = HighCapacity (4)
18EFF417h	0 2 D 39h 05h ; . Write IsoProfile_Runtime = Disturbed (5)
18EFF417h	0 2 D 39h 06h ; . Write IsoProfile_Runtime = Service (6)
18EFF417h	0 2 D 39h 07h ; . Write IsoProfile_Runtime = UG Profile (7)
18EFF417h	0 2 D 3Bh 01h ; . Write IsoProfile_Startup = StandardFastStartup (1)
18EFF417h	0 2 D 3Bh 02h ; . Write IsoProfile_Startup = Standard (2)
18EFF417h	0 2 D 3Bh 05h ; . Write IsoProfile_Startup = Disturbed (5)
18EFF417h	0 2 D 3Bh 06h ; . Write IsoProfile_Startup = Service (6)
18EFF417h	0 2 D 3Bh 07h ; . Write IsoProfile_Startup = UG Profile (7)
18EFF417h	0 2 D 57h 01h ; . Write StartSelfDiagnostic = 1 (Offline)
18EFF417h	0 2 D 57h 02h ; . Write StartSelfDiagnostic = 2 (Offline+Communication)
18EFF417h	0 2 D 71h 00h ; . Write EarthLift = connect
18EFF417h	0 3 D 47h 64h 00h ; . Write Isolation Threshold_Error = 100kOhm
18EFF417h	0 3 D 49h 00h 00h ; . Write Isolation Threshold_Timeout_Measurement = 0s (off)
18EFF417h	0 3 D 49h 01h 00h ; . Write Isolation Threshold_Timeout_Measurement = 1s
18EFF417h	0 3 D 49h 3Ch 00h ; . Write Isolation Threshold_Timeout_Measurement = 60s
18EFF417h	0 3 D 4Bh B8h 0Bh ; . Write Isolation Threshold_Warning = 3000kOhm
18EFF417h	0 3 D 59h 00h 00h ; .Write Selftest Period = 0s (off)
18EFF417h	0 3 D 59h 01h 00h ; .Write Selftest Period = 10s (1x10s)
18EFF417h	0 3 D 67h 00h 00h ; . Write VoltThreshold = 0 (off)

18EFF417h 0 3 D 67h 0Ah 00h ; . Write VoltThreshold = 10V